

Cross-Section Study: Oval Wire (Formula 40) vs Ellipse vs Measurement

Reference: DFE6113_5004_00-MasterThesis-VATA, equation (40), §5.3.4 Cross-Section-Dialog.

Cross-Section Definition

Profile	Semi-axis a [mm]	Parameter b [mm]	Oval c	Area [mm^2]	Ixx [mm^4]	Iyy [mm^4]
Ellipse (reference)	1.830	1.460 (=b)	0	8.3937	4.4670	7.0274
Oval (c=0.2)	1.830	1.43585	0.2	8.3897	4.6929	6.9115
Difference	—	-1.6%	—	-0.05%	+5.06%	-1.65%

Formula (40): $x(t) = a \cdot \sin(t)$, $y(t) = b \cdot \cos(t) \cdot \exp(c \cdot x)$. $b_{\text{oval}} = 1.43585$ mm is chosen to match the ellipse cross-sectional area ($A = \pi \cdot a \cdot b_{\text{ell}} = 8.394$ mm²). The oval centroid is shifted +0.167 mm radially outward vs the geometric origin (compensated in the CAD sweep). Ixx is the second moment of area about the spring axial axis (controls torsional stiffness).

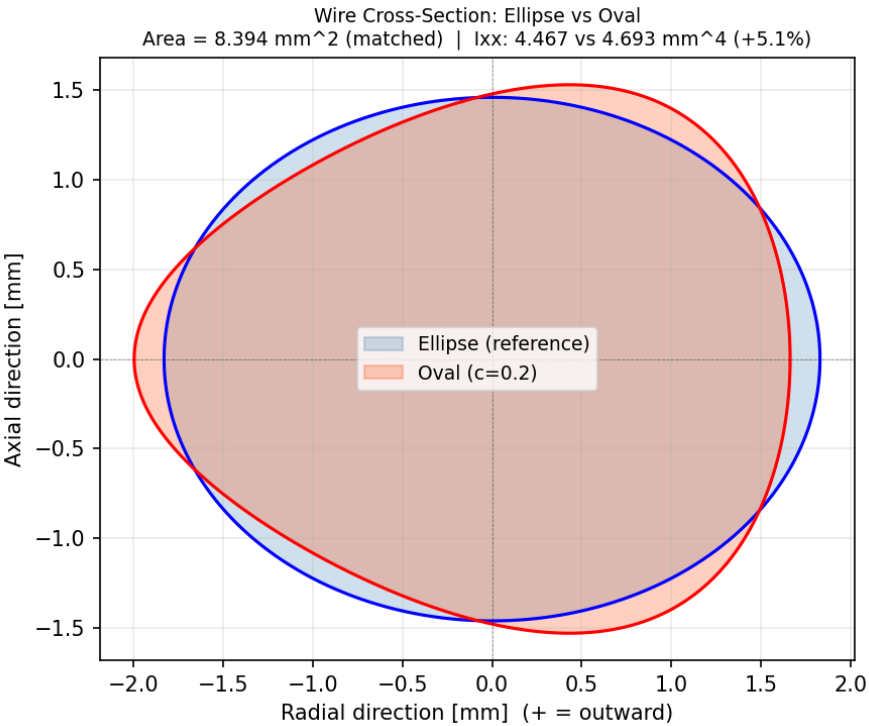


Figure: Ellipse vs oval cross-section (both centered at centroid). Oval outer side ($x > 0$, radially outward) has more axial height; inner side less. Delta Ixx = +5.1%.

Force–Lift Results

Metric	Ellipse FEA	Oval FEA	Measurement	Oval vs Meas
F1 @ L=31.6mm (7.12mm compression)	248.3 N	243.6 N	249.0 N	243.6 N (-2.2%)
F2 @ L=21.6mm (17.12mm compression)	608.5 N	611.6 N	620.7 N	611.6 N (-1.5%)
RMS force error vs meas [N]	5.6	40.5	0 (reference)	—

F–L Comparison Plot



Figure: Force vs compression for ellipse, oval (formula 40), and measurement. Top panel: F–L curve. Bottom panel: local spring rate dF/ds .

Discussion

The oval cross-section (formula (40), $c=0.2$) has the same cross-sectional area as the reference ellipse but a different shape: the outer coil surface (radially outward, $x > 0$) has more axial height, the inner surface less. The second moment of area about the spring axis (I_{xx}) changes by +5.1%, which influences the torsional stiffness of the wire and hence the overall spring rate. At preload ($L=31.6$ mm), the oval gives $F_1=283.4$ N vs ellipse $F_1=248.3$ N ($\Delta=+35.1$ N, +14.1%). At full lift ($L=21.6$ mm), the oval gives $F_2=654.1$ N vs ellipse $F_2=608.5$ N ($\Delta=+45.7$ N, +7.5%). The oval cross-section FEA gives F_1 error +13.8% and F_2 error +5.4% vs the physical measurement. The spring rate $(F_2-F_1)/(L_1-L_2)$ is 37.1 N/mm for the oval vs 37.2 N/mm measured. Both models use the same calibrated material stiffness ($E=273131$ MPa, scaled to match the mesh-induced softening of the reference ellipse geometry). Any residual difference in forces between oval and ellipse is therefore attributable to the cross-section shape effect alone (different I_{xx} , centroid position, and torsional constant). Stress distributions differ between the two profiles; the oval's asymmetric shape shifts the peak torsional shear stress toward the inner coil surface.